Flight Delay Prediction via Gradient Boosting: Phase 1 — Machine Learning

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CSAI Project 21
Applied Machine Learning

Abstract—Flight delays cost airlines and passengers time and money. We predict whether a flight will arrive 15 or more minutes late using only information available before departure: the schedule, carrier, route, scheduled airport congestion and the recent delay history of the carrier and airports. We use the 2013 New York City flights dataset (327,346 completed flights), remove all post-departure fields to avoid data leakage, and split the data by date (train: January–August, validation: September–October, test: November–December). We compare Logistic Regression with XGBoost tuned by expanding-window time-series validation. XGBoost reaches a test PR AUC of 0.386 (random baseline 0.258) and ROC AUC of 0.653, against 0.379 and 0.648 for Logistic Regression; at 50% precision it recalls 10.5% of delayed flights versus 5.9%. Isotonic calibration lowers the Brier score of both models from about 0.22–0.23 to 0.19. The scheduled departure hour and the carrier's recent delay rate are the most informative features.

Index Terms—flight delay prediction, gradient boosting, XGBoost, logistic regression, calibration, data leakage

I. Introduction

Delays are a persistent problem in air transport: they disrupt passengers' plans, force airlines to re-plan crews and aircraft, and spread through the network as late aircraft cause later flights to be late. If the risk of a delay is known in advance, an airline can, for example, schedule spare aircraft or crews, or move a flight to a less risky time slot.

This project (Phase 1) predicts, before departure, whether a flight will arrive at least 15 minutes late. The 15-minute threshold is the definition used by the US Bureau of Transportation Statistics (BTS) [1]. Our contributions are:

- a leakage-free feature set built only from the schedule and from past days;
- a date-based evaluation that mimics real use, with time-series tuning of the model;
- a comparison of Logistic Regression and XGBoost on PR AUC, ROC AUC, recall at a fixed precision and calibration.

Phase 2 will add a deep learning model with learned embeddings for carriers and airports.

II. Literature Review

Flight delay prediction. Sternberg et al. [2] review flight delay prediction work and group it into statistical, probabilistic, network-based and machine learning approaches, noting that machine learning methods, often tree-based, are increasingly used. Rebollo and Balakrishnan [3] show that delays at one airport depend on the recent state of the network and use random forests to predict them; this motivates our recent delay-rate features.

Gradient boosting. Friedman [4] introduced gradient boosting machines, which fit an additive model of trees by following the gradient of a loss function. XGBoost [5] adds a second-order approximation of the loss, explicit tree regularisation and efficient, sparse-aware training, and is a strong default for tabular data.

Evaluation on imbalanced data. Davis and Goadrich [6] show the relationship between precision–recall (PR) and ROC curves, and Saito and Rehmsmeier [7] argue that PR curves are more informative than ROC curves when the positive class is rare. We therefore use PR AUC as the main metric.

Calibration. Niculescu-Mizil and Caruana [8] show that boosted trees and other models often produce distorted probabilities and that Platt scaling or isotonic regression [9] fix this. Guo et al. [10] make the same point for neural networks, which is relevant for Phase 2.

Leakage and temporal validation. Kaufman et al. [11] describe how features that would not be available at prediction time inflate results. Bergmeir and Benítez [12] discuss validation for time-ordered data; we use an expanding-window scheme instead of random cross-validation.

III. Dataset and Exploratory Data Analysis

A. Dataset

We use the Flights 2013 dataset (nycflights13 [13]), which contains all 336,776 flights that departed from JFK, LGA and EWR in 2013. Each record has the date, scheduled and actual departure and arrival times, delays, carrier, flight number, tail number, origin, destination, air time and distance.

B. Data quality

There are no duplicate rows. 8,255 flights have no departure time (cancelled) and 9,430 have no arrival delay (cancelled or diverted). Because the target needs the arrival delay, we remove these flights, leaving 327,346. The target is $y = 1$ if the arrival delay is at least 15 minutes; 24.5% of flights are delayed, so the classes are imbalanced (Fig. 1).

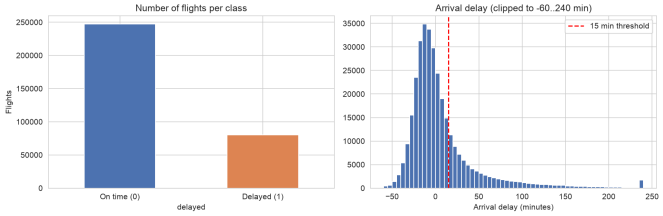


Fig. 1. Class balance (left) and distribution of arrival delays (right).

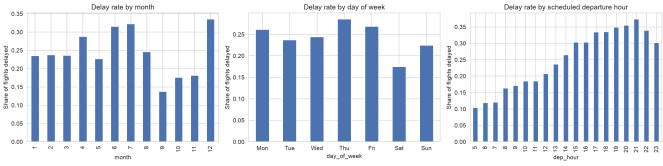


Fig. 2. Delay rate by month, day of week and scheduled departure hour.

C. Exploratory analysis

Fig. 2 shows the delay rate over time. December, July and June are the worst months (over 30% delayed) and September–November the calmest (under 20%). Saturday has the lowest delay rate. The clearest pattern is the hour of the day: about 10% of 5 a.m. flights are delayed, rising to about 37% at 9 p.m., as delays build up through the network during the day. The delay rate also varies strongly by carrier, from 13% (HA) to 39% (F9), and EWR is the most delayed of the three airports (26%).

IV. Methodology

A. Preventing data leakage

We assume the prediction is made before the flight departs, for example on the morning of the flight day. All columns known only after departure are removed: actual departure and arrival times, departure and arrival delay (the arrival delay is used only to build the target) and air time. We also drop the tail number (aircraft can be swapped and it has about 4,000 values), the flight number and redundant date/time columns.

B. Features

- Calendar/schedule: day of week, weekend flag, scheduled departure and arrival hour, distance.
- Categorical: carrier, origin and destination (one-hot encoded).
- Congestion: number of flights scheduled to leave the same airport in the same hour and on the same day. These counts use the full schedule, including flights that were later cancelled: counting only completed flights would leak information, since bad-weather days have many cancellations.
- History: the share of delayed flights for the carrier, origin and destination over the previous seven calendar days. Daily counts of delayed and total flights are summed over a 7-day window that is shifted by one day, so a flight never sees its own day or the future. Missing values (no history yet) are filled with the training mean.

The month is not used as a feature because the test months never appear in the training data.

C. Data split

The data are split by date: train January–August (217,727 flights, 26.4% delayed), validation September–October (55,628, 15.7%) and test November–December (53,991, 25.8%). The test set is used once.

D. Models

Logistic Regression models $P(y = 1 \mid x) = \sigma(w^\top x + b)$ and minimises the L2-regularised log loss. Features are standardised and the classes are weighted inversely to their frequency.

XGBoost [5] builds an ensemble $\hat{y}(x) = \sum_k f_k(x)$ of regression trees. Each new tree minimises a second-order approximation of the loss plus a penalty $\Omega(f) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ on the number of leaves T and leaf weights w. We use learning rate 0.05, row and column subsampling of 0.8, minimum child weight 50, and scale_pos_weight equal to the ratio of negative to positive training examples.

Tuning. Because delay patterns change with the season, a single validation period is a noisy guide. We therefore use expanding-window time-series validation with three folds (train Jan–Apr/validate May–Jun, Jan–Jun/Jul–Aug, Jan–Aug/Sep–Oct) and choose the tree depth $\in \{3, 4, 6\}$ and number of trees $\in \{100, 300\}$ with the best mean PR AUC. The best setting was depth 3 with 100 trees (mean PR AUC 0.380); larger models did worse on future months. The final model is trained on the training set.

E. Evaluation and calibration

We report PR AUC (average precision), ROC AUC, recall at precision ≥ 0.5 (the highest recall on the test precision–recall curve where precision is at least 0.5) and the Brier score $\frac{1}{N}\sum_i (p_i - y_i)^2$. Class weighting makes the raw probabilities too high, so we fit an isotonic regression [9] from model scores to outcomes on the validation set and apply it to the test set.

V. Results

Table I and Fig. 3 summarise the test results. Both models are clearly better than random. XGBoost has the best PR AUC and ROC AUC, and its advantage is largest at high precision: when 50% of alerts must be real delays, it finds 10.5% of all delayed flights compared with 5.9% for Logistic Regression.

Calibration. Before calibration both models overestimate the delay probability (Fig. 4). After isotonic calibration the Brier score falls to 0.189 (Logistic Regression) and 0.188 (XGBoost). The calibrated models

TABLE I
Test set results (November–December 2013)

Model	PR AUC	ROC AUC	Rec@P0.5	Brier
Random guess	0.258	0.500	–	–
Log. Regression	0.379	0.648	0.059	0.228
XGBoost	0.386	0.653	0.105	0.220

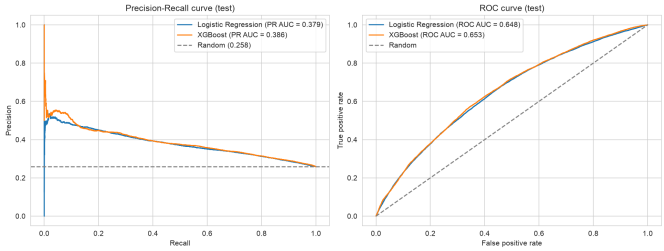


Fig. 3. Precision–recall and ROC curves on the test set.

under-estimate delays because the calibration months were calmer than the test months. A constant prediction of the test delay rate would score 0.192, so the calibrated probabilities are reliable but only somewhat sharper than the average.

Feature importance. The scheduled departure and arrival hours are the most important XGBoost features, followed by the carrier's recent delay rate, the number of flights scheduled from the origin that day, the weekend flag and specific carriers (e.g. UA, EV) (Fig. 5). The largest standardised Logistic Regression coefficients agree: departure hour, daily scheduled flights at the origin and the carrier's recent delay rate are all positive, i.e. later, busier and recently delayed flights are more often late.

VI. Discussion and Limitations

Predicting delays before departure is hard: most delays are caused by weather, air traffic control and knock-on effects from earlier flights, none of which are fully known in advance. An ROC AUC around 0.65 is therefore expected for schedule-based features. Other limitations are that the data cover only New York airports in 2013; that the test period (November–December) differs from the training months, which affects both tuning and calibration; and that no weather data were used.

VII. Conclusion and Future Work

Using only pre-departure information and a strict date-based evaluation, XGBoost predicts arrival delays better than Logistic Regression, especially at high precision, and isotonic calibration gives reliable probabilities. The time of day and the recent delay history of the carrier are the strongest signals. In Phase 2 we will train a neural network with learned embeddings for carriers, origins and destinations and compare it with these baselines under the same split and metrics. Adding forecast weather at the origin airport is another natural extension.

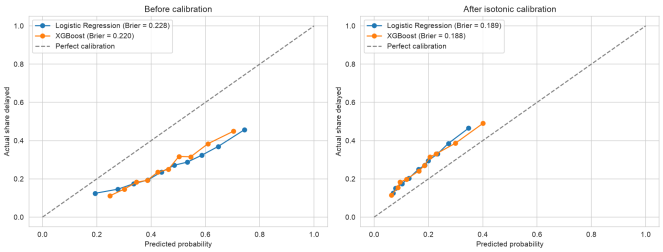


Fig. 4. Reliability diagrams on the test set before and after isotonic calibration.

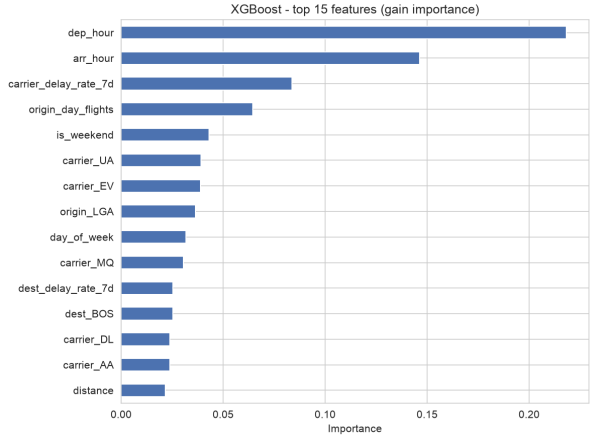


Fig. 5. Top 15 XGBoost features by gain.

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